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Classical vs. Semantic Communications for IoT Occupancy Detection

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Abstract

Wireless sensor networks often transmit large amounts of data over channels affected by noise, creating a trade-off between communication efficiency and reliable information delivery. This project investigates whether semantic communication can reduce the amount of transmitted data while preserving performance on a downstream task. Using the UCI Occupancy Detection dataset, we developed and compared two wireless communication systems: a classical system that transmits quantized sensor measurements and reconstructs the original data, and a semantic system that compresses the measurements into a representation designed to preserve information relevant to occupancy prediction. Both systems were evaluated under varying signal-to-noise ratios (SNRs), using BPSK, QPSK, and 16-QAM modulation, along with repetition, Hamming(7,4), and convolutional channel coding with Viterbi decoding. The semantic system transmitted 32 bits per sample, compared with 64 bits for the classical system, reducing the transmitted bit count by 50%. At low SNRs, semantic communication achieved approximately 69–80% occupancy prediction accuracy, compared with 47–55% for classical communication across the tested configurations. At high SNRs, however, classical communication reached approximately 98% accuracy, while semantic communication achieved 88–89%. These results demonstrate a trade-off between communication efficiency and prediction accuracy. Semantic communication can reduce transmission overhead while maintaining useful task performance, particularly under noisy channel conditions. However, compression also removes information that may be useful for prediction. The effectiveness of this approach therefore depends on the accuracy requirements and communication constraints of the intended application.

Contents

1 Introduction **2**
1.1 Related Work 2

2 System Model **4**

3 Proposed Method **4**
3.1 Classical Communication System 4
3.2 Semantic Communication System 5

4 Results **6**
4.1 Ablation Study 8

5 Discussion **9**

6 Conclusion **9**

7 Future Work **10**

1 Introduction

The increasing use of artificial intelligence and Internet of Things (IoT) devices has created a growing demand for wireless communication. Applications such as smart thermostats, occupancy sensors, environmental monitoring systems, wearable devices, and industrial networks rely on the continuous exchange of data to support automated decisions. As the number of connected devices increases, transmitting and processing these large amounts of data can place greater demands on bandwidth, energy, and communication resources.

Traditional communication systems, known as classical communication systems, aim to reproduce the transmitted information as accurately as possible. Every bit is treated as important, regardless of its meaning or relevance to the task being performed. Although this approach is necessary when exact data reconstruction is required, it can introduce unnecessary communication overhead when the receiver only needs information for a specific purpose.

Semantic communication offers an alternative approach. Rather than prioritizing the exact reconstruction of every transmitted bit, semantic communication focuses on preserving the meaning or task-relevant information contained in the original data. By compressing information into a representation that supports a particular downstream task, a semantic system may reduce the number of bits that need to be transmitted.

This project investigates the trade-off between classical and semantic communication using an occupancy-detection task. The two systems are evaluated under the same noisy wireless channel conditions to determine how reducing transmitted information affects bit error rate (BER) and occupancy prediction accuracy.

Problem Description

IoT devices often collect environmental measurements continuously. For example, an occupancy-detection system may use temperature, humidity, light, and carbon dioxide (CO2) measurements to determine whether a room is occupied.

The UCI Occupancy Detection dataset contains these four environmental features along with an occupancy label. The measurements contain patterns that can help distinguish occupied from unoccupied conditions. In particular, light and CO2 levels can provide useful information about changes in room occupancy, while temperature and humidity may provide additional context.

A classical communication system transmits the original sensor measurements so that they can be reconstructed at the receiver. However, if the only objective is to predict occupancy, transmitting every measurement with the same precision may not be necessary. A semantic system could instead learn a compact representation that prioritizes the information most useful for occupancy prediction.

The central question of this project is whether transmitting fewer bits can still provide enough information to complete the same downstream task.

1.1 Related Work

A. **Classical Communications** Classical communication systems are designed to transmit information from a sender to a receiver with as few bit errors as possible. A typical system begins by converting source information into a sequence of bits. Channel coding then adds redundancy to help detect or correct errors introduced during transmission. The coded bits are mapped onto a modulation scheme and transmitted through a communication channel. At the receiver, demodulation and channel decoding are used to recover the original information.

This approach focuses on accurately reproducing the transmitted bit sequence, regardless of what the information represents. As a result, transmitting large amounts of data can require

considerable bandwidth and energy, especially when some of the original information is not essential to the receiver's final task [3].

B. **Semantic Communications**

In 1948, Claude Shannon introduced a mathematical framework for understanding the transmission of information over communication channels. His work focused primarily on the technical problem of transmitting symbols accurately. Shannon and Weaver also discussed the semantic and effectiveness levels of communication, which consider the meaning of transmitted information and its influence on the receiver's actions [1].

Semantic communication revisits these broader aspects of communication. Instead of requiring the receiver to reconstruct every original bit, a semantic system aims to preserve the information necessary to understand a message or complete a particular task.

Recent approaches use deep learning to create an encoder-decoder system. The encoder transforms the original input into a compact latent representation, while the decoder uses the received representation to recover information relevant to the intended task. This project applies this approach to environmental sensor measurements, with occupancy prediction as the downstream objective [2].

C. **Channel Coding**

Channel coding introduces redundancy into transmitted data to improve reliability when errors occur. This project evaluates three channel-coding methods. Repetition coding transmits each bit multiple times. The receiver uses majority voting to determine the original bit, allowing the system to recover from some transmission errors at the cost of additional transmitted symbols. Hamming(7,4) coding converts every four data bits into seven coded bits. This adds redundancy while allowing the receiver to correct single-bit errors within each codeword [4]. Convolutional coding uses a sliding memory structure to encode the input bit sequence. The Viterbi algorithm then identifies the most likely transmitted sequence based on the received data [5]. These coding methods provide different trade-offs between error correction, transmission overhead, and implementation complexity.

D. **AWGN Channel**

The additive white Gaussian noise (AWGN) channel is a standard model for evaluating communication systems. It represents a transmitted signal that has been affected by additive noise with a Gaussian distribution. The signal-to-noise ratio (SNR) measures the signal power relative to the noise power and is commonly expressed in decibels (dB). A higher SNR indicates that the signal is stronger relative to the noise, while a lower SNR indicates that noise has a greater effect on the received signal. Using an AWGN channel provides a controlled environment for comparing communication systems under different noise conditions. In this project, the channel is used to evaluate how changes in SNR affect bit transmission and occupancy prediction.

E. **Digital Modulation Techniques** Digital modulation converts coded bits into signals that can be transmitted through a communication channel. This project evaluates three modulation schemes. Binary phase-shift keying (BPSK) represents one bit per symbol using two possible phase states. Because the two states are relatively easy to distinguish, BPSK generally provides strong resistance to noise. Quadrature phase-shift keying (QPSK) represents two bits per symbol using four possible phase states. It can transmit more bits per symbol than BPSK at the same symbol rate. 16-ary quadrature amplitude modulation (16-QAM) represents four bits per symbol using sixteen different signal states. Although this increases the number of bits transmitted per symbol, the signal states are more closely spaced, making the system more sensitive to noise. These modulation schemes demonstrate the trade-off between transmission efficiency and noise tolerance [3].

2 System Model

This project compares classical and semantic communication systems using the same environmental sensor data and a shared AWGN channel model. Both systems receive measurements from the UCI Occupancy Detection dataset and transmit information to a receiver, where the received data is used to predict whether a room is occupied.

In the classical system, the four sensor measurements are quantized into bits and transmitted through the communication channel. At the receiver, the measurements are reconstructed and used for occupancy prediction.

In the semantic system, a learned encoder first compresses the four sensor measurements into a lower-dimensional representation. The resulting latent values are quantized and transmitted through the same type of communication channel. A decoder then processes the received representation to produce information for occupancy prediction.

Both systems are evaluated using different modulation schemes, channel-coding methods, and SNR values. This shared setup allows their transmission reliability and downstream prediction performance to be compared under different channel conditions.

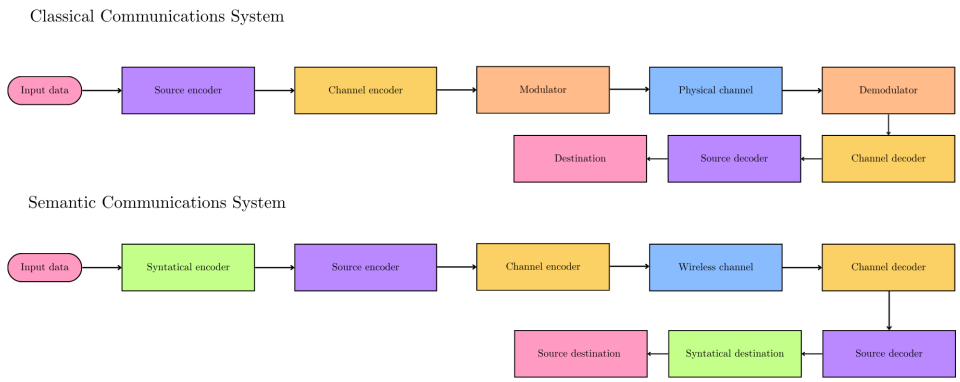


Figure 1: End-to-end block diagram of the classical and semantic pipelines.

3 Proposed Method

3.1 Classical Communication System

The classical communication system transmits the original environmental sensor measurements. Each sample contains four features: temperature, humidity, light, and CO2. Each feature is quantized into 16 bits, resulting in a total of 64 bits per sample. The resulting bit sequence is passed through a channel encoder. Depending on the experiment, the system uses repetition coding, Hamming(7,4) coding, or convolutional coding. The coded bits are then mapped onto a modulation scheme, such as BPSK, QPSK, or 16-QAM, before being transmitted through the AWGN channel. At the receiver, the signal is demodulated and channel-decoded. The recovered bits are then dequantized to reconstruct the four sensor measurements. These reconstructed measurements are

Semantic Dimension	Bits per Sample	Compression Ratio
1	8	8:1
2	16	4:1
4	32	2:1
8	64	1:1
16	128	No compression (expansion)

Figure 2: Transmitted bits per sample and compression ratio at different semantic dimensions.

used to predict occupancy. Because the classical system attempts to preserve the original sensor values, its performance depends on how accurately the transmitted bits can be recovered.

3.2 Semantic Communication System

The semantic communication system compresses the original sensor measurements into a learned representation before transmission. Rather than transmitting every sensor value independently, the encoder transforms the four input features into a compact latent representation intended to preserve information relevant to occupancy prediction. The model was trained for 200 epochs using the Adam optimizer with a learning rate of 0.01. The training data came from the UCI Occupancy Detection dataset, which contains approximately 8,143 samples with four environmental sensor features and an occupancy label. The semantic representation used in the main experiments had a dimension of four. Each latent value was represented using eight bits, producing a total of 32 bits per sample. In comparison, the classical system transmitted 64 bits per sample by representing each of the four original features with 16 bits. This reduced the number of transmitted bits by 50%, requiring only half as many bits per sample as the classical system.

The two communication systems differ in the type of information they prioritize. The classical system attempts to reconstruct the original sensor measurements, while the semantic system focuses on preserving information that supports occupancy prediction. This distinction affects how transmission errors influence the final task. In the classical system, errors in the transmitted bits can change the reconstructed sensor measurements. In the semantic system, the received latent representation is processed by a decoder trained for the occupancy-prediction task. As a result, the semantic system may retain useful prediction performance even when the received representation is imperfect.

However, semantic compression also has limitations. Reducing the number of transmitted bits can remove information that would otherwise help the receiver make more accurate predictions. The semantic system therefore involves a trade-off between communication efficiency and the amount of information available for the downstream task.

The two approaches also differ in computational requirements. The classical pipeline primarily involves quantization, channel coding, modulation, demodulation, and decoding. The semantic pipeline introduces neural-network encoding and decoding, which require additional computation. Although semantic communication reduces the number of transmitted bits in this experiment, its overall computational cost and end-to-end latency were not measured.

The choice between classical and semantic communication depends on the requirements of the application. Classical communication is preferable when exact reconstruction is essential, such as in applications where every transmitted value must be preserved. Semantic communication may be useful when the receiver primarily needs information to complete a particular task, such as occupancy detection or environmental monitoring. In these situations, transmitting a smaller amount of task-relevant information may be worthwhile if the resulting prediction accuracy meets the application's requirements.

4 Results

The experiments evaluate the two communication systems under different SNR conditions, modulation schemes, and channel-coding methods. Their performance is examined using bit error rate and occupancy prediction accuracy.

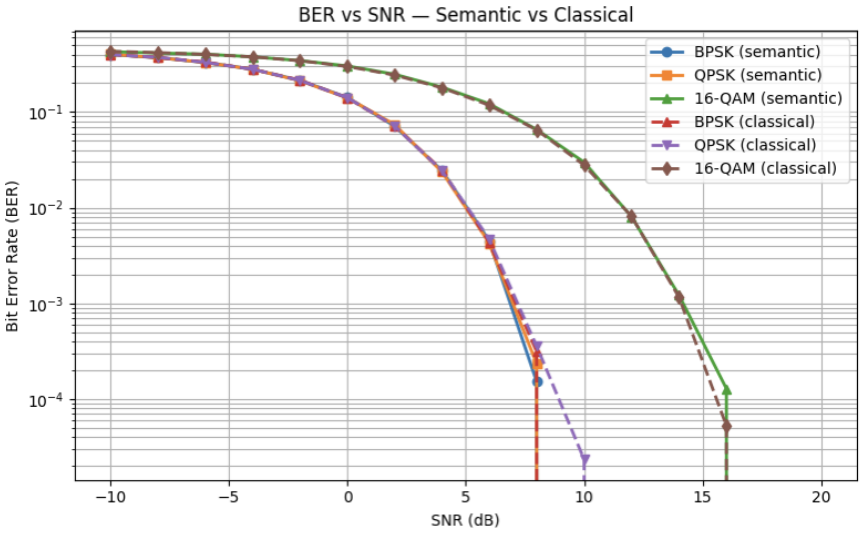


Figure 3: BER versus SNR for the evaluated coding and modulation schemes.

Bit Error Rate

Figure 3 shows that the BER curves for the classical and semantic systems are essentially identical when the same modulation scheme and channel conditions are used. This is expected because the physical-layer bit error rate depends on the modulation and channel rather than the meaning of the transmitted bits.

For BPSK and QPSK, the BER curves overlap closely and fall below (10^{-4}) at approximately 8 dB SNR. In comparison, 16-QAM requires approximately 8 dB more SNR to reach a similar error rate. This difference reflects the greater sensitivity of 16-QAM to noise because its signal states are more closely spaced.

Although the two communication systems experience similar physical-layer bit error rates, their occupancy prediction accuracies differ. This suggests that the way information is represented and processed can influence the effect of transmission errors on the final task.

Occupancy Prediction Accuracy

Figure 4 compares the performance and characteristics of classical and semantic communication.

At low SNRs, particularly around -10 dB, the semantic system achieves approximately 70% occupancy prediction accuracy, compared with approximately 47–55% for the classical system. This suggests that the semantic representation can preserve useful information for occupancy prediction even when the communication channel introduces substantial noise.

As the SNR increases beyond approximately 4–6 dB, the performance of the two systems begins to separate. Classical communication continues improving toward approximately 98% accuracy, while semantic communication plateaus around 88–89%.

The difference at high SNR reflects the limitations of compression. The classical system reconstructs the original sensor measurements using their 16-bit representations, while the semantic system transmits a smaller latent representation. Although the latent representation preserves

Dimension	Classical Communication	Semantic Communication
Bits per sample	64 (16 bits x 4 features)	32 (8 bits x semantic dim. 4)
Design goal	Exact bit-level reconstruction	Preserve task-relevant meaning
Low-SNR accuracy (-10 to 0 dB)	~47-55%	~69-80%
High-SNR ceiling (>=10 dB)	~98% (noiseless classifier bound)	~88-89% (latent-code capacity bound)
Robustness to bit errors	Errors directly corrupt sensor values	Decoder tolerant of residual bit errors
Computational cost	Lower (quantize, code, modulate)	Higher (neural encoder/decoder + training)
Best suited for	Exact-recovery tasks (records, audits, safety alarms)	Bandwidth/power-constrained IoT inference tasks

Figure 4: Comparison of classical and semantic communication performance and characteristics.

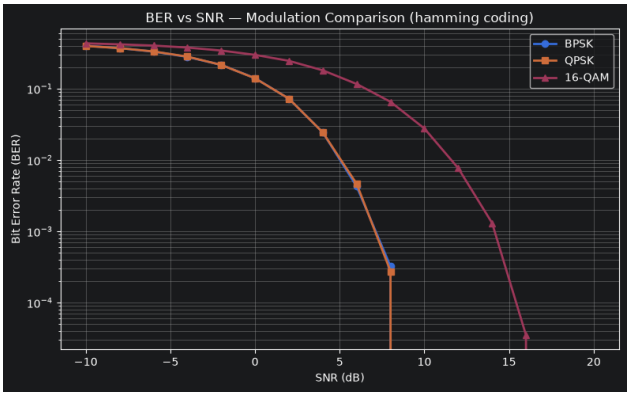


Figure 5: BER versus SNR for BPSK, QPSK, and 16-QAM under Hamming(7,4) channel coding.

information useful for occupancy prediction, it does not retain every detail of the original measurements.

Modulation Comparison

The modulation experiments compare BPSK, QPSK, and 16-QAM while keeping Hamming(7,4) channel coding fixed. The results show that BPSK and QPSK achieve similar BER and occupancy prediction performance. In contrast, 16-QAM requires approximately 8 dB higher SNR to achieve comparable performance.

This difference illustrates the trade-off between the number of bits transmitted per symbol and resistance to noise. Although 16-QAM transmits more bits per symbol, its closely spaced signal states make it more vulnerable to errors. BPSK and QPSK provide greater noise tolerance because their signal states are more distinguishable.

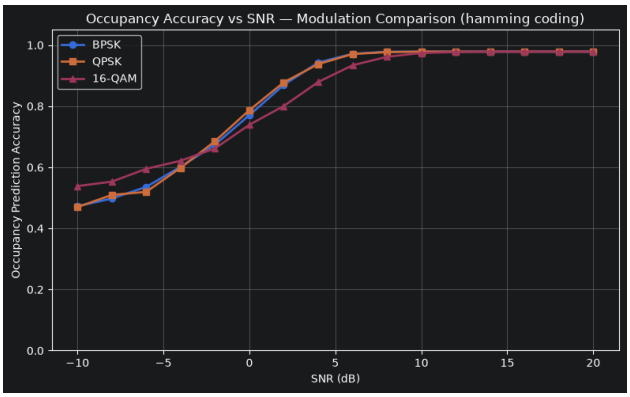


Figure 6: Occupancy prediction accuracy versus SNR for BPSK, QPSK, and 16-QAM under Hamming(7,4) channel coding.

4.1 Ablation Study

To understand how individual design choices affect system performance, we varied the modulation scheme and channel-coding method while keeping the other experimental settings fixed.

First, we compared BPSK, QPSK, and 16-QAM while keeping Hamming(7,4) channel coding fixed. Figures 5 and 6 show the corresponding BER and occupancy prediction accuracy results. BPSK and QPSK achieved similar performance, while 16-QAM required approximately 8 dB higher SNR to reach comparable results. This occurs because 16-QAM represents more bits per symbol using more closely spaced signal states, making it more sensitive to noise.

Next, we compared repetition coding, Hamming(7,4), and convolutional coding with Viterbi decoding while keeping BPSK modulation fixed. Repetition coding achieved usable occupancy accuracy at lower SNRs in the tested configurations, while Hamming and convolutional coding approached similar performance at higher SNRs.

Repetition coding improves resistance to transmission errors by sending each bit multiple times and using majority voting at the receiver. However, this approach requires additional transmitted symbols. Hamming and convolutional coding introduce structured redundancy to correct errors with different levels of coding complexity. At higher SNRs, where fewer transmission errors occur, the relative benefit of additional error correction becomes smaller.

Finally, we compared the classical and semantic communication systems. The classical system transmitted 64 bits per sample, while the semantic system transmitted 32 bits per sample, reducing the transmitted bit count by 50%. At high SNRs, classical communication achieved approximately 98% occupancy accuracy, compared with 88–89% for semantic communication. At low SNRs, semantic communication achieved approximately 69–80% accuracy across the tested configurations, compared with 47–55% for classical communication.

These results demonstrate that the two systems prioritize different objectives. Classical communication preserves more information from the original sensor measurements, allowing it to achieve higher accuracy when transmission conditions are favorable. Semantic communication reduces transmission overhead by compressing the input into a representation designed for the occupancy-prediction task. This can be advantageous under noisy conditions, but the compression also removes information that may be useful for prediction.

Overall, the ablation study demonstrates that modulation, channel coding, and the communication approach each influence system performance. Modulation affects the balance between transmission efficiency and noise tolerance, channel coding affects error correction, and the choice

between classical and semantic communication determines how much information is transmitted and preserved for the downstream task.

5 Discussion

The results demonstrate a trade-off between communication efficiency and occupancy prediction accuracy. Compared with the classical communication system, the semantic system transmitted approximately half as many bits per sample while achieving 88–89

Although semantic communication did not achieve the same high-SNR accuracy, it maintained useful prediction performance while reducing the amount of transmitted information. This suggests that transmitting every bit of the original data may not always be necessary when the receiver's objective is to complete a specific task.

The difference between the two systems becomes especially important when considering noisy channel conditions. At low SNRs, semantic communication achieved higher occupancy prediction accuracy than classical communication in the tested configurations. Because the semantic encoder is trained to preserve information relevant to occupancy prediction, its compressed representation may retain useful task information even when transmission errors occur.

However, the lower high-SNR accuracy also demonstrates the limitations of semantic compression. Reducing the amount of transmitted information means that some details from the original sensor measurements are not preserved. These details may still be useful for making more accurate predictions. Therefore, the effectiveness of semantic communication depends on identifying which information is essential to the task and determining how much accuracy can reasonably be sacrificed to reduce communication costs.

These findings have potential applications in wireless sensor networks, particularly in environmental monitoring. Sensors used to monitor conditions such as water quality, temperature, and dissolved oxygen may need to transmit data continuously while operating under limited power and bandwidth. A semantic communication system could potentially reduce transmission overhead by prioritizing meaningful changes or patterns in environmental conditions.

The acceptable trade-off between efficiency and accuracy would depend on the application. A small reduction in prediction accuracy might be acceptable when monitoring general environmental trends, but less acceptable when detecting sudden changes that require immediate attention. Further testing would be necessary to determine whether the approach performs reliably with real-world environmental sensors.

Overall, this project demonstrated that designing an effective communication system involves more than maximizing transmission accuracy. Engineers must also consider the purpose of the transmitted information, the resources available, and the consequences of prediction errors.

6 Conclusion

This project investigated whether semantic communication can reduce the amount of transmitted information while maintaining useful performance on the same downstream task as classical communication. Using the UCI Occupancy Detection dataset, we compared two systems under different channel conditions, modulation schemes, and channel-coding methods.

The semantic system transmitted 32 bits per sample, compared with 64 bits for the classical system, reducing the transmitted bit count by 50%. At high SNRs, classical communication achieved approximately 98% occupancy prediction accuracy, while semantic communication

achieved 88–89%. At low SNRs, semantic communication maintained higher occupancy prediction accuracy in the tested configurations.

These findings show that transmitting fewer bits can still provide enough information to complete a task with useful accuracy. However, reducing the transmitted information also means losing details that may contribute to more accurate predictions. The choice between classical and semantic communication therefore depends on the application's requirements and the acceptable balance between communication efficiency and prediction accuracy.

The experiments also demonstrated that modulation and channel coding influence communication reliability. BPSK and QPSK achieved similar performance, while 16-QAM required higher SNR to achieve comparable results. Repetition, Hamming, and convolutional coding also presented different trade-offs between error correction and transmission overhead.

7 Future Work

Future work could investigate improved semantic representations that preserve more predictive information without substantially increasing the number of transmitted bits. Additional experiments could also evaluate different channel conditions, measure energy consumption and end-to-end latency, and test the system using real-world wireless sensor networks.

One potential application is environmental monitoring. Low-power sensors could use semantic communication to transmit meaningful changes in water quality or other environmental conditions while reducing communication overhead. For example, instead of continuously transmitting every sensor measurement, a system could prioritize information that indicates meaningful changes in environmental conditions.

Ultimately, the goal is not simply to transmit fewer bits, but to identify the minimum amount of information necessary to accomplish a task reliably.

References

[1] C. E. Shannon, "A mathematical theory of communication," *Bell System Technical Journal*, vol. 27, pp. 379–423, 1948.

[2] Z. Qin, Y. Tao, Z. Lu *et al.*, "Semantic communications: Principles and challenges," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 1, pp. 1–30, 2023.

[3] J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed. McGraw-Hill, 2008.

[4] R. W. Hamming, "Error detecting and error correcting codes," *Bell System Technical Journal*, vol. 29, no. 2, pp. 147–160, 1950.

[5] A. J. Viterbi, "Error bounds for convolutional codes and an asymptotically optimum decoding algorithm," *IEEE Transactions on Information Theory*, vol. 13, no. 2, pp. 260–269, 1967.

[6] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.

[7] L. Candanedo, "Occupancy detection," 2016. [Online]. Available: https://doi.org/10.24432/C5X01N